

A variational approach to progressive lens design

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The ability of the human eye to accommodate (i.e. to see both near and far objects sharply) decreases with age. In order to correct this defect, so-called progressive-addition lenses (PALs) have been developed. This paper presents a new method for the design and quality control of PALs. The desired properties of the PAL are combined in an error functional, involving curvature properties of the front surface of the lens. It is shown how to efficiently minimize this error functional iteratively and we prove quadratic convergence. We also present an efficient method for reconstructing the geometry of a manufactured PAL based on sensor data obtained by measuring devices. The combination of geometry reconstruction and optimization provides a useful tool for the main design loop of PALs. The system has been developed and implemented in close cooperation with the optical industry. © 1998 Elsevier Science Ltd. All rights reserved.

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INTRODUCTION

In order to see objects at different distances sharply, the lens of the human eye is able to alter its curvature and thus its power of refraction. Our ability to accommodate, i.e. to change the shape of this lens, decreases with age, due to gradual stiffening of the lens. To properly correct this defect, known as *presbyopia*, we need glasses with low refractive power for viewing objects at far distances, and higher power for near distances. Thus, the required accommodation effort of the human eye is (partially) supplied by an increasing refractive power of the spectacle lens. Such a glass is called a *multi-focal lens* or a *progressive-addition lens* (PAL). Usually, a PAL has a large *far vision area* with low refractive power in the upper part, and a smaller *near vision area* with higher power in the lower part. Moreover, the distribution of refractive power on the PAL should be smooth and monotonic.

The first PAL was manufactured in 1967 by Maitenaz¹. His approach is based on simple heuristics (versus exact computation) and makes use of only a few free parameters. Another approach to PAL design computes a main curve

with increasing mean curvature on the front surface. The whole surface is then defined by a conic which is moved along the main curve, also having increasing mean curvature from top to bottom. Free form surfaces have been used in a patent specification by Zeiss². However, rather than considering the surface as a whole, this technique only takes into account the location of some sample points when computing the PAL. When minimizing the errors at the sample points, common numerical optimization methods are used, which require large computational effort. In Ref. ³ a variational approach is outlined. In contrast to our approach the functional to be minimized does not take into account the exact optical quantities. It rather consists of six different terms, each being derived using simple heuristics.

This paper describes a new method for determining the geometry of PALs, which enables the PAL designer to create a variety of different designs, and minimizes optical errors as much as possible. The computation takes into account the exact optical quantities of a lens and specific requirements of the designer (e.g. size and location of the near and far vision area). In order to detect manufacturing errors and to validate that tolerances for optical parameters are not exceeded, the geometry of a measured PAL has to be reconstructed. Sophisticated methods for general reverse engineering problems usually require long computation times. Therefore we developed a fast method which can handle noisy data and produces surfaces in the standard tensor product B-spline format.

The method described in the following has been developed and implemented in close cooperation with the optical industry. The results shown in the paper demonstrate the benefits of our approaches in practical applications. Moreover, lenses manufactured according to the method described in this paper have proven to show a better error distribution than currently available progressive lenses.

The paper is organized as follows. We begin with an introduction of the relevant optical factors and derive an appropriate error functional for the surface of a PAL. In Section 3 we present algorithms for minimizing this functional, while the reconstruction method is described in Section 4.

THE OPTICS OF PALS

Characteristics of ophthalmic lenses

In ophthalmic lens optics, the characteristic quantity of a lens is its *vertex power*. Consider a lens with rotation axis l , as in Figure 1. l intersects the lens' back face at the *back vertex B*.

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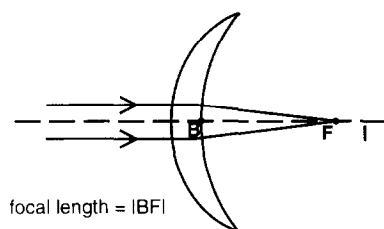


Figure 1 Focal length of rotationally symmetric converging lens

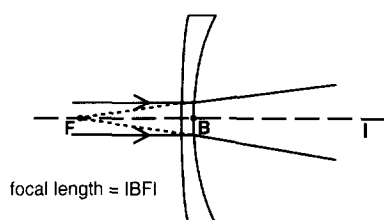


Figure 2 Focal length of rotationally symmetric diverging lens

An incoming parallel bundle of rays along l intersects l at the *focus* F . The lens' *focal length* is $|BF|$, the distance from B to F . Finally, the reciprocal value $P = 1/|BF|$ is called the (*vertex*) *power*. It is specified in *dioptr* = *dpt* (m^{-1}). Note that for a diverging lens, the focus F will be virtual, being positioned in front of the lens (see Figure 2). Therefore $|BF|$ is understood to be directed, being negative in the case of diverging lenses (see also Ref. ⁴).

The refractive power of a lens is given by the *thick lens formula*

$$P = (1 - n)\kappa^b + \frac{(n - 1)\kappa^f}{1 - d(1 - \frac{1}{n})\kappa^f} \quad (1)$$

where κ^f and κ^b are the curvatures of the front and back surface respectively, and d denotes the thickness of the lens, n being the index of refraction. Assuming a negligible thickness, we obtain the *thin lens formula*:

$$P = (n - 1)(\kappa^f - \kappa^b) \quad (2)$$

In the case that one of the surfaces, e.g. the front surface, is non-symmetrical, there are two different principal curvatures and eqns (1) and (2) have to be applied separately for each principal curvature direction. In that case, we get two refractive powers P_1 and P_2 . The quantity $A := |P_1 - P_2|$ is called *astigmatism*. A non-vanishing astigmatism results in two focal lines at different distances. Obviously, a (symmetrical) eye, using a lens with $A \neq 0$, cannot accommodate on both focal lines. We require $A = 0$ for spectacle lenses*.

Visual defects

An eye is said to be correctly sighted (emmetropic) if a point at infinity gets mapped sharply on the retina without any accommodation effort being made, i.e. when the lens is unaccommodated. In optical terms, a thin bundle of parallel light rays is deflected so as to converge to the same point on the retina (as in Figure 3(a)). We call an eye *short sighted* (*far sighted*), if an incoming parallel ray bundle has its point

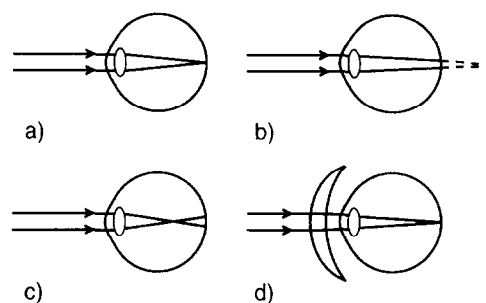


Figure 3 (a) An eye without visual defects; (b) a far-sighted eye; (c) a short-sighted eye; (d) a far-sighted eye with correction glass

of intersection in front of (behind) the retina (see Figure 3(b) and 3(c)).

These defects are corrected by spectacle lenses with negative or positive power of refraction respectively (see Figure 3(d)).

When looking at an object at a finite distance, the eye has to accommodate, i.e. the lens has to bend in order to increase its curvature, and thus its power of refraction. With increasing age the eye loses part of its accommodative ability; it becomes *presbyopic*. In this situation unifocal spectacles can only correct within a certain viewing range. Using a PAL with low refractive power in the upper part (far vision area) and higher refractive power in the lower part (near vision area), the wearer can compensate the missing accommodation ability by moving the eye downward on the lens. An additional short or far sightedness is compensated as well, using a back face of appropriate curvature according to eqn (2).

Requirements on progressive lenses

Due to the thin lens formula (eqn (2)) we are able to separate the front and back faces of a lens. Assuming a spherical back face, we can therefore translate the requirements on a PAL into requirements on the curvature properties of an ideal front surface.

- (1) A certain mean curvature $P_f = (\kappa_1 + \kappa_2)/2$ is specified in the far vision area. In optical terms, the mean curvature is called (*surface*) *power*. Actually P_f could be chosen arbitrarily. However, flat or highly curved glasses are usually not desired, because of aesthetic reasons and due to optical errors caused by oblique astigmatism. Therefore P_f is often chosen in the range of 3–5 dpt†.
- (2) The required surface power P_n in the near vision area is not specified directly. Instead, the progression $P_n - P_f > 0$ corresponding to the amount of missing accommodation will be specified.
- (3) Ideally, the front surface should be 'locally spherical', i.e. $C := |\kappa_1 - \kappa_2|$ should vanish. In ophthalmology, $|\kappa_1 - \kappa_2|$ is called (*surface*) *astigmatism* or *cylinder*.

Unfortunately, a surface being 'locally spherical' (i.e. having vanishing astigmatism) has to be part of a sphere, i.e. it cannot have varying mean curvature (see also Ref. ⁵ for a quantitative statement). Consequently, we have to compute a surface which fulfils eqns (1) and (2) and shows minimal astigmatism $|\kappa_1 - \kappa_2|$. Usually, one tries to get minimal astigmatism at the far and near vision

* Currently, a possible astigmatic defect of a human eye will be corrected by an appropriate toric back surface of the lens.

† Adjustment to the necessary refractive power is achieved by an appropriate back surface of the lens.

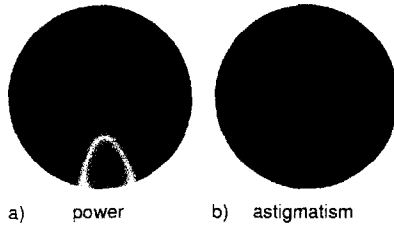


Figure 4 (a) Refractive power. (b) Optical error (astigmatism). Blue indicates lowest values for refractive power (a) and optical errors (b) respectively, red indicates maximal values

areas and on a *zone of progression* which connects both areas. The non-avoidable astigmatism should be reduced as far as possible and be placed near the boundary of the lens.

Figure 4 shows the front face of a typical multi-focal lens. The refractive power is displayed color encoded on the left. Blue indicates lower, red higher values of power. The power increases smoothly from the upper far vision area to the lower near vision area. Optical errors (astigmatism) are shown on the right of Figure 4. Errors at the far and near vision areas and along the main viewing line are very small. Figure 5 shows the variation of mean curvature along the main viewing line. In the lower near vision part, the main viewing line is usually displaced into the nasal region of the lens, in order to take into account that the eyes have to converge when looking at shorter distances.

Characterization of a PAL by an error functional

Representing the front surface of a lens by the graph of a function $f : \Omega \rightarrow \mathbb{R}$, with domain $\Omega \subset \mathbb{R}^2$ chosen as a disc, we can formulate the task of computing an appropriate front surface of the lens as a variational problem for the function f .

- The desired distribution of mean curvature on the graph of f is specified by a function $P : \Omega \rightarrow \mathbb{R}$.
- We specify weight functions $\alpha, \beta, \Omega \rightarrow \mathbb{R}$, $\alpha, \beta > 0$ for penalizing surface astigmatism and deviation of the prescribed mean curvature respectively.
- The quality of the surface is measured by the following error functional J :

$$J(f) := \int_{\Omega} \alpha(\kappa_1 - \kappa_2)^2 + \beta \left(\frac{\kappa_1 + \kappa_2}{2} - P \right)^2 dS \quad (3)$$

(We measure the error in the L^2 sense, since this allows an easier numerical treatment than, for example, the maximum norm.)

- The lens corresponding to the specified design parameters P , α and β is the solution to the following optimization problem:

Determine f such that $J(f)$ is minimal

The design parameters α , β and P are functions defined on the disc. The mean curvature function P is usually chosen to be constant along horizontal lines, and increases continuously from P_f to P_n . The function α specifies how much an occurrence of astigmatism is penalized by the functional. Similarly, β weights the deviation from the required mean curvature. Thus the designer specifies higher values of α and β in near and far vision areas as well as along the main viewing line, whereas at the other parts lower values are used. Thus, the weight functions α and β provide good

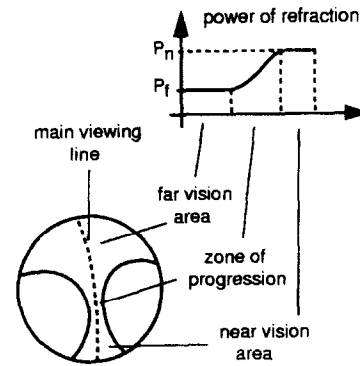


Figure 5 Distribution of refractive power on a PAL

control over the optical quantities of the resulting lens (see Figure 6). In Section 3.6 we describe in more detail, how the weight functions can be specified.

MINIMIZING THE ERROR FUNCTIONAL

In the last section, we introduced the functional

$$J(f) = \int_{\Omega} \alpha(\kappa_1 - \kappa_2)^2 + \beta \left(\frac{\kappa_1 + \kappa_2}{2} - P \right)^2 dS$$

This functional is well-defined for functional surfaces having continuous partial derivatives up to the order of two. There is no hope of finding an analytic expression of the solution in terms of α , β and P for the minimum of this functional. Only a numeric solution can be evaluated. We use a Ritz–Galerkin approach for the numerical analysis. Thus we consider a finite dimensional subspace V of $C^2(\Omega)$, e.g. V could be the space of bicubic tensor product B-splines. Then, we have to determine $f^* \in V$ such that $J(f^*) \leq J(f)$ holds for every $f \in V$.

Minimizing highly non-linear functionals like J using conventional numerical methods, e.g. the conjugate gradient method, is very time consuming^{6,7}. On the other hand, it is relatively easy to minimize a quadratic functional. In fact, minimizing a quadratic functional can be reduced to solving a linear system of equations. Therefore our strategy is to find local approximations to J that are quadratic. In addition, by iteration we will improve the approximations of J using quadratic functionals step by step. A similar method is described in Ref. ⁸ where quadratic functionals are used to minimize certain curvature functionals.

The idea is as follows: let us assume that we can find a family of quadratic functionals J_f , one for each *reference surface* f , such that $J_f(f) = J(f)$ holds. If the mapping $(f, g) \mapsto J_f(g)$ is continuous, it follows that J_f is a good local approximation to J in a neighborhood of f . We can therefore perform the following,

Algorithm 3.1

- (1) Determine a rough approximation f^0 to f^* .
- (2) Do until convergence: Determine f^{n+1} by minimizing J_{f^n} .

If the approximating functionals J_f are chosen carefully, the sequence converges: $f^n \rightarrow f^*$ for $n \rightarrow \infty$. Now we want to discuss two quadratic approximations that turn out to be appropriate for our lens-design functional J .

Approximation with Taylor series

Since J is only dependent on the principal curvatures of the graph of f , only the first- and second-order derivatives of f have an influence on J . Thus, we can write J as

$$J(f) = \int_{\Omega} K(f_u, f_v, f_{uu}, f_{uv}, f_{vv}) \, du \, dv$$

with $K: \mathbb{R}^5 \rightarrow \mathbb{R}$. To determine K some elementary differential geometry is necessary: the mean curvature $H = (\kappa_1 + \kappa_2)/2$ and the Gaussian curvature $G = \kappa_1 \kappa_2$ of the graph of f are given by

$$H = \frac{1}{2} \frac{(1 + f_u^2)f_{vv} - 2f_{uv}f_{uv} + (1 + f_v^2)f_{uu}}{g^3}$$

$$\text{and } G = \frac{f_{uu}f_{vv} - f_{uv}^2}{g^4} \quad (4)$$

with area element $g = \sqrt{1 + f_u^2 + f_v^2}$. The relation $(\kappa_1 - \kappa_2)^2 = 4(H^2 - G)$ is also easily verified. Thus, for the functional J given in eqn (3) K is given by

$$K(f_u, f_v, f_{uu}, f_{uv}, f_{vv}) = (4\alpha(H^2 - G) + \beta(H - P)^2)g \quad (5)$$

Inserting the expressions for H and G of eqn (4) into eqn (5) leads to an explicit representation of K in terms of $f_u, f_v, f_{uu}, f_{uv}, f_{vv}$. For simplicity we use $L(f)$ as an abbreviation for $(f_u, f_v, f_{uu}, f_{uv}, f_{vv})$. Then we can write

$$J(f) = \int K(L(f)).$$

In order to achieve a quadratic approximation of J in the neighborhood of a certain reference function f^0 , we use the Taylor expansion of K at f^0 , omitting the terms of degree higher than two. So the quadratic approximation of J is given by

$$J_{f^0}(f) = J(f^0) + \int \text{grad } K(L(f^0))^t L(h) + \frac{1}{2} \int L(h)^t \text{Hess } K(L(f^0)) L(h) \quad (6)$$

where $h := f - f^0$ and $\text{grad}K$ and $\text{Hess}K$ denote the gradient and the Hessian of K respectively. Obviously

$J_{f^0}(f) = J_{f^0}(f^0 + h)$ is a quadratic functional in h and can therefore be minimized efficiently.

The following theorem shows that this approximation is well suited for minimization purposes.

Theorem 3.2

Let f^* be an isolated non-degenerated minimum of J , i.e. $d/(dt)J(f^* + tg)|_{t=0} = 0$ and $d^2/(dt^2)J(f^* + tg)|_{t=0} > 0$ holds for every $g \in V$. Then Algorithm 3.1 produces a sequence $(f^n)_{n \in \mathbb{N}}$ that converges quadratically to f^* for every $f^0 \in V$ sufficiently close to f^* .

Proof. A necessary condition for f^* to be an extremal value of J is that the Fréchet derivative of J at f^* vanishes, i.e. $D_{f^*}J : g \mapsto [D_{f^*}J](g)$ has to be the zero functional (see, for example, Ref. 9). Moreover, the assumption $d^2/(dt^2)J(f^* + tg)|_{t=0} > 0$ for all $g \in V$ assures that f^* is an isolated zero of $DJ : f \mapsto 4D_fJ$. Therefore, it is sufficient to show that Algorithm 3.1 leads to a quadratically convergent sequence $f^n \rightarrow f^*$ for $n \rightarrow \infty$, where f^* is a zero of DJ . In fact, we will show that Algorithm 3.1 is equivalent to the well-known Newton iteration, applied to $DJ : f \mapsto D_fJ$.

In order to do so, we briefly remark that for a quadratic functional $Q(f) = Q_0 + Q_1(f) + Q_2(f, f)$ the minimum f^* of Q can be characterized by $2Q_2(f^*, f) = -Q_1(f)$ for all f . Equivalently, $f^* = -1/2T^{-1}(Q_1)$ where $T : V \rightarrow \mathcal{L}(V) : f \mapsto Q_2(f, \cdot)$ assigns to every $f \in V$ the linear form $Q_2(f, \cdot) : g \mapsto Q_2(f, g)$.

The Newton iteration for solving $DJ = 0$ is

$$f^{n+1} := f^n - T_{f^n}^{-1}(D_{f^n}J) \quad (7)$$

where T_{f^n} is the operator as above corresponding to $Q_2 = D_{f^n}^2J$. In order for T_{f^n} to be invertible and for the Newton iteration to have quadratic convergence $D_{f^n}^2J$ has to be positive definite for f^n sufficiently close to f^* . Since this condition is fulfilled by the assumption $d^2/(dt^2)J(f^* + tg)|_{t=0} > 0$ for all $g \in V$, the Newton iteration fulfils our purposes.

On the other hand, one iteration step in Algorithm 3.1 consists of finding the minimum f^{n+1} of $J_{f^n}(f) = J_{f^n}(f^n + h^n) = J(f^n) + [D_{f^n}J](h^n) + 1/2[D_{f^n}^2J](h^n, h^n)$ with $h^n := f - f^n$.

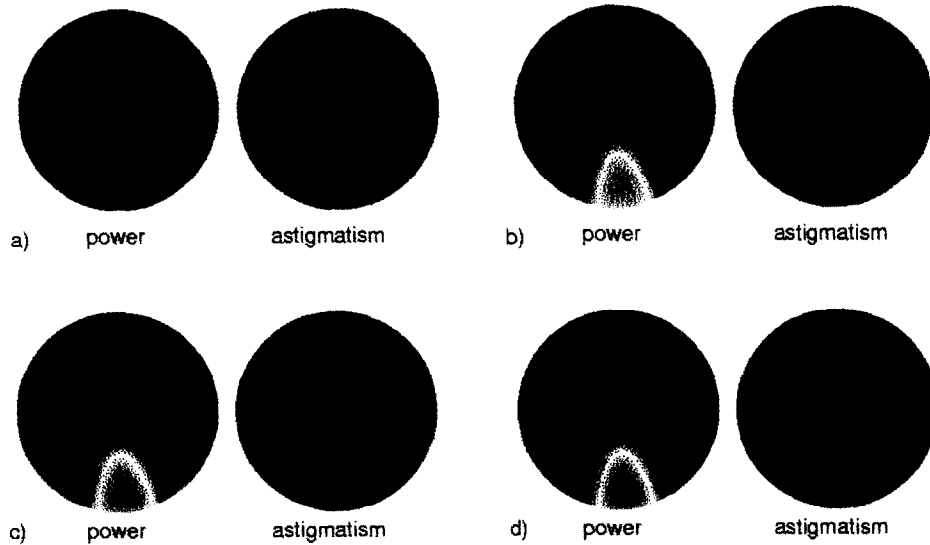


Figure 6 (a) Initial surface (sphere). (b), (c) and (d) show three iteration steps of the method of frozen first derivatives

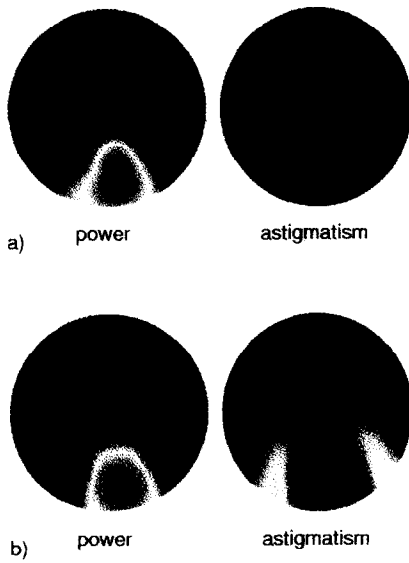


Figure 7 Examples of currently available PALs. Numerous defects are visible

The above remark shows that

$$h^n = -\frac{1}{2} \left(\frac{1}{2} T_{f^n} \right)^{-1} (D_{f^n} J) = -T_{f^n}^{-1} (D_{f^n} J) \quad (8)$$

and thus $f^{n+1} = f^n + h^n = f^n - T_{f^n}^{-1} (D_{f^n} J)$; this is exactly one iteration step in eqn (7). ■

Approximation with frozen first-order derivatives

Minimizing a quadratic functional $Q(f) = Q_0 + Q_1(f) + Q_2(f, f)$ with linear $Q_1(f)$ and bilinear $Q_2(f, f)$ in a finite dimensional space $V = \text{span}(B_i)$ is equivalent to solving a linear system of equations, where the corresponding matrix is computed by evaluating $Q_2(B_i, B_j)$ for every pair of basis functions (B_i, B_j) . Using the method described above, we have $Q_2(B_i, B_j) = 1/2 \int L(B_i)^t \text{Hess } K(L(f^0)) L(B_j)$ which can be determined by numerical integration. The evaluation of $L(B_i)$, $L(B_j)$ and $\text{Hess } K(L(f^0))$ can be done in a preprocessing step for every knot (u_p, v_q) appearing in the quadrature formula. Thus, the most time-consuming step is the computation of the products $L(B_i)^t \text{Hess } K(L(f^0)) L(B_j)$. Since $\text{Hess } K(L(f^0))$ is a 5×5 matrix, this requires 54 floating point operations. With the minimizing method described below, we work with 3×3 matrices instead, reducing the computation effort to only 24 floating point operations.

If we take a closer look at the curvature function K given in Eq. (3.1), we observe that K is already quadratic in its arguments f_{uu}, f_{uv}, f_{vv} , or more exactly

$$K(f_u, f_v, f_{uu}, f_{uv}, f_{vv}) = c(f_u, f_v) + r(f_u, f_v)^t \hat{L}(f) + \hat{L}(f)^t A(f_u, f_v) \hat{L}(f) \quad (9)$$

where $c(f_u, f_v) \in \mathbb{R}$, $r(f_u, f_v) \in \mathbb{R}^3$ and $A(f_u, f_v) \in \mathbb{R}^{3 \times 3}$ only depend on the first-order derivatives of f , $A(f_u, f_v)$ being a symmetric matrix. $\hat{L}(f)$ is used as an abbreviation for (f_{uu}, f_{uv}, f_{vv}) . We omit representing c , r and A explicitly in terms of f_u, f_v . This can easily be done using the formulae for K given in Section 3.1.

The second observation is that for ‘flat’ surfaces, i.e. for

functions f with small first-order derivatives f_u, f_v , the curvature function K depends primarily on the second order derivatives f_{uu}, f_{uv}, f_{vv} , or, more precisely,

$$\begin{aligned} \frac{dK}{df_u}(0, 0, f_{uu}, f_{uv}, f_{vv}) &= \frac{dK}{df_v}(0, 0, f_{uu}, f_{uv}, f_{vv}) \\ &= 0 \text{ for all } f_{uu}, f_{uv}, f_{vv} \in \mathbb{R} \end{aligned} \quad (10)$$

which is easily verified. Since spectacle lenses are indeed rather flat, it seems appropriate to use as an approximation to J a curvature functional that only depends on the second-order derivatives. Such a functional, with respect to a reference surface f^0 , is given by

$$J_{f^0}(f) := \int c(f_u^0, f_v^0) + r(f_u^0, f_v^0)^t \hat{L}(f) + \hat{L}(f)^t A(f_u^0, f_v^0) \hat{L}(f) \quad (11)$$

which is obviously quadratic and fulfils $J_f(f) = J(f)$ for all $f \in V$.

When computing the matrix of the corresponding linear equation, we have to integrate $\int \hat{L}(B_i)^t A(f_u^0, f_v^0) \hat{L}(B_j)$ numerically for all basis functions B_i and B_j . In contrast to the approximation method described in Eq. (3.1), this involves only 3×3 matrix arithmetic.

Discussion of optimization methods

We found that both methods lead to a fast converging sequence of approximative solutions. Because the second method is more than two times faster than the Taylor-series approach, it will always be used in the early design stage and for lenses with low refractive power. However, if lenses with very high power have to be designed, the solution f^* cannot be expected to represent a ‘flat’ surface, so the method we introduced first is preferable. In Figure 7 three steps of the iteration are shown. The method of frozen first derivatives has been used.

The initial surface in Figure 6(a) is a sphere with the same curvature as the far vision area of the desired front surface. After the first step, the final result is almost reached; only the surface power in the near vision area is not yet sufficient (Figure 6(b)). The differences in Figure 6(c) and 6(d) are quite immaterial, i.e. the sequence becomes quasi-stationary after the second iteration step.

Obviously, both minimizing methods require a sufficiently good approximation f^0 to the desired surface f^* , because only then we can expect convergence. In all cases we investigated, it proved to be sufficient to simply use a flat disc $f^0 \equiv 0$ as the initial function. In order to reach the minimum faster, a sphere of radius $2/(P_f + P_n)$ can be used. Figures 5, and 8 show different PAL designs that have been computed using the method of frozen first derivatives. All PALs have a progression of 2 dpt. The computation took 5 s on a SGI MIPS R10000 195 MHz processor. The solution is given as a 30×30 tensor product B-spline patch, trimmed to a disc. In order to compare the results of the optimization method to currently available lenses we displayed power and astigmatism of two progressive addition lenses in Figure 7. The optical characteristics of these lenses are 3 dpt for the far vision area and 5 dpt for the near vision area.

The lens displayed in Figure 7(a) has tolerable astigmatism, but the refractive power does not increase monotonically from the upper to the lower part; the usable portion of

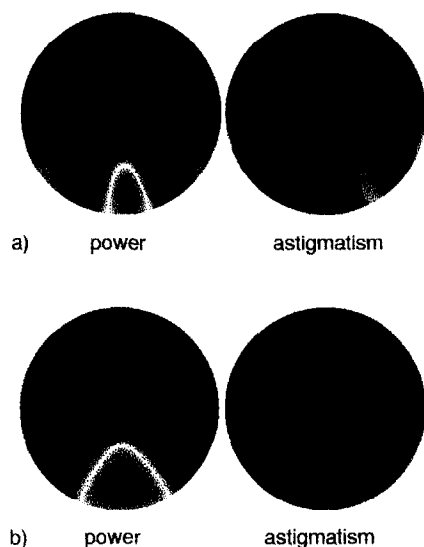


Figure 8 Multi-focal lenses, computed with the method introduced here

the far vision area is therefore restricted to a small region. In contrast, the lens in *Figure 7(b)* has correct refractive powers along the whole main viewing line. However, intolerable deviations occur next to the near vision area. Moreover, the techniques used to produce this lens could not sufficiently reduce the astigmatism. The lenses of *Figures 4, and 8* obviously do not have these problems: the variation of the refractive power from far to near vision area is monotonic and the astigmatism is reduced to a satisfactory level.

Implementation

When performing the optimization as described above, we first have to choose a representation of the functional surface $f : \Omega \rightarrow \mathbb{R}$, i.e. a linear function space V and a basis of V have to be specified. It turns out that bicubic tensor product B-splines are a good choice. The basis is given by the functions $B_i(u)B_j(v)$ where B_i denotes the normalized, univariate B-spline basis function over an equidistant knot vector. A functional surface $f \in V$ is then characterized by its *control points* b_{ij} , i.e. $f(u, v) = \sum b_{ij}B_i(u)B_j(v)$. Concerning definition, notation and basic results on B-splines we refer to Ref. ¹⁰ or Ref. ¹¹. Cubic tensor product B-spline surfaces are well-suited for our purposes, because:

- cubic splines provide C^2 continuity, i.e. the resulting curvature distribution will be continuous;
- cubic B-spline basis functions have minimal support among cubic polynomial spline bases, which is important for efficiency reasons and numerical stability;
- tensor product surfaces allow computations to be done efficiently. For example, we will interpolate data over a regular grid and this can be done very efficiently in a tensor product setting. In order to ensure existence and uniqueness of a numerical solution, certain constraints have to be added. For our problem, it turns out to be sufficient to require $f(0, 0) = f_u(0, 0) = f_v(0, 0) = 0$. These constraints can easily be integrated into the optimization process. In fact, the constraints simply result in restricting J to a linear subspace V' of V . Thus, J is minimized over V' instead of V .

So far we have not mentioned how the weight functions α

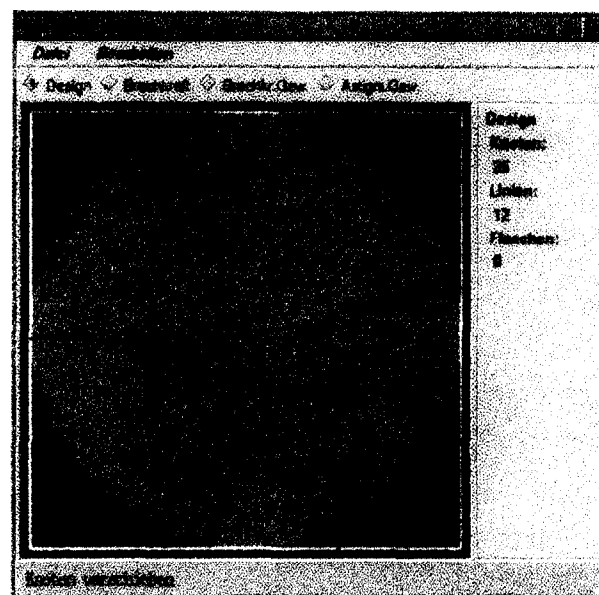


Figure 9 A typical partitioning of a lens

and β are actually specified. In our concrete implementation, the user subdivides the lens into areas, where different weights are necessary (see *Figure 9*). The boundary curves are cubic B-splines and can be drawn and manipulated interactively using common graphical input devices. On each area, the weights are set to a constant value, by clicking the area and adjusting a slider. This allows easy adjustment of a chosen design. An augmentation of the far vision area, for example, can be done by adjusting its boundary curve. Further subdivision can be done if the initial partitioning has been shown to be too coarse. Using an area-sampled average of α and β , we can also provide continuous weight functions if required. However, we found, that piecewise constant functions are sufficient.

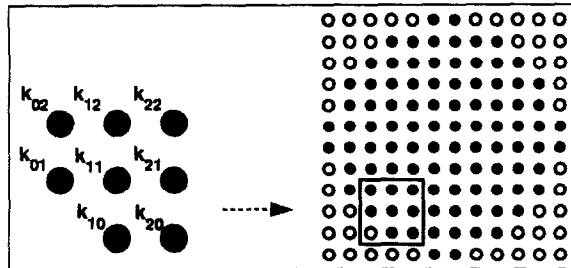
QUALITY CONTROL AND VISUALIZATION OF PALS

In the previous sections we have described a method for constructing PALS. However, the developed framework can also be used for inspection and quality control. After a PAL has been computed, the optical power has to be carefully checked over the whole glass. If considerable deviations from the desired result are discovered, the design parameters are adjusted, and the optimization process is started again. Another important aspect for commercial use of the design tool is the possibility to compare the quality of new designed lenses and existing lenses (for which geometric models may not exist). We therefore need appropriate reverse engineering methods for PALS. We present a method that is both efficient and sufficiently accurate. Four tasks have to be performed.

- (1) The existing lens is sampled on its front surface to obtain a representation as a functional surface $f : \Omega \rightarrow \mathbb{R}$. This is done by mechanical measuring devices. Usually, grids of size 30×30 to 60×60 are used.
- (2) Since tensor product surfaces have a rectangular parameter domain, we extrapolate the discrete measurement data to the bounding rectangle of Ω (see below).

Table 1 Mask coefficients k_{ij} for the least-squares method (left) and the differential equation method (right)

$k_{0,2} = -1/7$	$k_{1,2} = -4/7$	$k_{2,2} = +5/7$	$k_{0,2} = -1$	$k_{1,2} = +2$	$k_{2,2} = -1$
$k_{0,1} = +8/7$	$k_{1,1} = -4/7$	$k_{2,1} = -4/7$	$k_{0,1} = +2$	$k_{1,1} = -4$	$k_{2,1} = +2$
	$k_{1,0} = +8/7$	$k_{2,0} = -1/7$		$k_{1,0} = +2$	$k_{2,0} = -1$

**Figure 10** A 3×3 mask is used to complete the data set

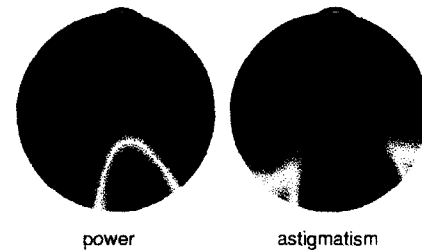
- (3) A low-pass filter is applied to the data in order to reduce noise caused by measurement errors. We use linear filtering with an appropriate triangular filter.
- (4) The gridded data are interpolated by a bicubic tensor product B-spline f .

Step (2) deserves some more explanation. We consider a 3×3 grid (u_i, v_j) , $i, j = 0, \dots, 2$ of data, on which the value for (u_0, v_0) is to be extrapolated by the other eight values (see Figure 10). We find a quadratic polynomial q that is the least-squares fit for the eight known values $f_{i,j} := f(u_i, v_j)$, and set $f_{0,0} := q(u_0, v_0)$. Alternatively, we can define $f_{0,0}$ by solving a discrete version of the differential equation $(\partial^4 f)/(\partial x^2 \partial y^2) = 0$. The idea behind both methods is that the curvature information given by the eight known values should be extrapolated to the missing knot. Both methods lead to a linear extrapolation formula of the form $f_{0,0} = \sum k_{i,j} f_{i,j}$. The coefficients $k_{i,j}$ for both the least-squares method and the differential equation method are given in Table 1.

This 3×3 mask can now be used to complete the whole set of measured data by applying the mask to every 3×3 sub-grid, where one corner value is missing.

Thus, the data set is computed step by step, processing from the interior to the borders. The computation time is negligible for both methods. We found that the first method produces better results for PALs, whereas the second one is slightly better for uni-focal lenses. Using the method described above, a regular set of measured data can be transformed easily into a geometric model. The computation time for geometry reconstruction is negligible, taking only fractions of a second. All pictures of manufactured PALs shown in this paper were produced this way. The optical quantities are usually visualized by plotting curvature for the lens' front surface, where the terms of main interest are the surface power and surface astigmatism. Alternatively, one can use wavefront tracing in order to visualize the exact optical quantities (see, for example, Ref. ¹²) or simulate the view on a 3D scene through the lens.

Quality control is important in checking whether the optical parameters lie in the expected range. However, the method described above is also used for detecting manufacture errors. Figure 11 shows refractive power and astigmatism of a manufactured lens with defect in the far vision area. The data reconstruction methods easily allow us to detect such defects.

**Figure 11** Detection of manufacture errors. The defect is detected through color plots of power and astigmatism

CONCLUSION

We have presented a new method for the design and quality control of progressive-addition lenses (PALs). Using the thin lens approximation it is possible to reduce the design of a PAL to the optimization of its front surface, the back surface being fixed. This fact is important not only during the design process, but perhaps even more from a manufacturing point of view, since front and back surface are typically manufactured by different companies.

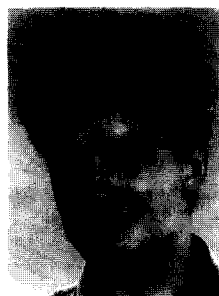
Optimization of the PAL's front surface is done by minimizing a suitably chosen error functional J involving only the principal curvatures of the front surface. We have presented a new efficient method to minimize this highly non-linear functional J by iteratively minimizing a sequence of quadratic approximations to J that are based on a reference surface. The construction of these quadratic approximations is based on a Taylor expansion of J , and we proved quadratic convergence for the iteration. Examples show that the sequence of surfaces becomes quasi-stationary after two iterations. Efficient functional minimization is essential in many Computer Aided Geometric Design applications. We plan to further investigate how our method can be applied in a more general context in the future.

Quality control is an important task in every manufacturing process. Controlling PALs requires efficient reverse engineering techniques. We have presented a problem-adapted reconstruction method that computes the geometry of a lens with negligible computation time.

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